LI, Ho Wa James

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Education

Hong Kong University of Science and Technology

Master of Philosophy (Mphil) in Bioengineering — Sept 2023 – Aug 2025

Supervisor: Prof. Henry Hei Ning LAM

Research Topic: SpectroVQ enables Efficient Information Compression and De-noising of Peptide Tandem Mass Spectrum using Deep Learning (Publication in progress) **CGPA:** 3.825

- Trained a Deep Vector Quantized model for efficient representation, compression and denoising of proteomics mass spectrometry data
- Experimenting and training variations of CNNs, Vision Transformers, VAEs, and Graph NNs on MS/MS proteomics data
- Analyzed on > 2 datasets and modified DDA proteomics data with existing pipelines (TPP, Alphapept)
- Collaborating on 2 other projects involving deep learning de novo protein sequencing and end-to-end proteomics analysis
- Supervising 2 FYP students on graph neural network-enhanced Spectral Library prediction and organism classification
- Setting up and Managing the lab's server including installing new hardware, partitions, softwares, pipelines and environments
- Teaching Assistant for Course BIEN 2310 Modeling for Chemical and Biological Engineering and BEHI 5007 Genomics, Proteomics and Metabolomics

Bachelor of Engineering (BEng) in Bioengineering — Sep 2019 – June 2023

CGPA: 3.508 **MCGA:** 3.598 (0.002 to First Class Honours)

- Undergraduate TA for course CENG 2210 Transport Phenomena I
- Course Highlights: Synthetic Biology, Data Science for Biology and Medicine, Deep Learning for Chemical and Biological Engineering

Selected Publications

SpectroVQ enables Efficient Information Compression and De-noising of Peptide Tandem Mass Spectrum using Deep Learning. — Feb 2025

James H.W. Li, Henry Lam.

73rd conference on Mass Spectrometry and Allied Topics American Society of Mass Spectrometry (ASMS) (Accepted and Presented)

Machine Learning-assisted Portable Electrical Impedance Tomography enables Accurate Lung Function Assessment — Feb 2025

James H.W. Li, Lawson M., Lui H., et. al.,

47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) (Accepted)

Machine Learning Enabled Portable Electrical Impedance Tomography System for Community Level Chronic Kidney Disease Classification and eGFR Estimation. — Feb 2025

James H.W. Li, Liu R., Lawson M.., et. al.,

47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) (Accepted)

Portable electrical impedance tomography (EIT) system stages nonalcoholic fatty liver disease for potential screening and monitoring at home. — Aug 2023

James H.W. Li, Adrien T., Fedi Z., et. al.,

45th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)

10.1109/EMBC40787.2023.10339955

Selected Experience

Part-time Research Internship, E-Sense, Hong Kong Centre for Cerebro-cardiovascular Health Engineering — June 2024 - Feb 2025

Topic: Improving algorithmic pipeline on Electrical Impedance Tomography (EIT) prediction on NAFLD and CKD
Supervisor: Dr. Russell Chan

- Reviewed and improved existing pipeline on NAFLD, CKD and FEV prediction based on collected clinical data
- Refactored and Implemented New backends for EIT scan acquisition, data storage and processing on Google Cloud Platform using docker and FastAPI
- Led 2 other summer interns to complete local machine learning pipelines and faulty detection algorithms

Research Internship and Co-op project (FYP), Gense Technologies Limited — Jan 2022 – Feb 2023

Topic: Develop machine learning algorithm for the prediction of NAFLD and CKD using anthropometric measures and Electrical Impedance Tomography (EIT) indicators
Supervisor: Dr. Russell Chan and Dr. Adrien Touboul

- Conducted clinical trials and validated EIT clinical data in Queen Mary Hospital
- Created aujtomatic pipeline to process collected inhouse and clinical data
- Explore, design, implement machine learning algorithms(random forest;logistic regression;SVM regression) on the prediction of NAFLD and CKD indicators using EIT scans

Undergraduate Research Opportunities Program, HKUST — June 2022 – Aug 2022

Supervisor: Prof. Ying Chau
Topic: Combinatorial peptides for Immunomodulation

- Performed RAW-BLUE Macrophage cell culture and solid-bead peptides synthesis
- Researched and reviewed on peptides targeting mechanism on TLRs and NF-kB signaling
- Designed and constructed a machine learning pipeline for prediction of macrophage response

SIGHT Project, Design for Global Health, HKUST — Feb 2022- Jun 2022

Topic: 餵你好, a modularized healthy food product for the young elderly

- Reviewed and Designed a modular diet lunchbox with tailored recipes targeted at young elderlies suffering from hypertension and malnutrition
- Introduced nutrientional concepts and engaged with young elderlies with our prototyped products at the Elderly Resources Centre for testing and receiving feedbacks from users

International Genetically Engineered Machine (iGEM), HKUST — Jan 2021 - Nov 2021

Topic: ShellBi, a Biosensor for Paralytic Shellfish Toxin

- Designed a Paralytic Shellfish Toxin biosensor with EnvZ-OmpR Two-Component System and Antisense scaffold modifications
- Modelled the systemic behaviour of the TCS, performed Molecular docking analysis using Autodock Vina and located putative binding proteins using phylogenetic analysis

Awards

Dean's List — June 2023

International Genetically Engineered Machine (iGEM) Gold Award — Jan 2021 - Nov 2021

Skills

Programming Languages: Python (Proficient), MATLAB (Proficient), Java, R, C++, UNIX programming

Machine learning frameworks: PCA, UMAP, t-SNE, KNN, linear regression, random forest, XGBoost, SVM regression, Multi-Layer Perceptron, Vision Models (VGG, ResNEt, Vision Transformer), Representation Learning (CLIP, SimCLR, Multi-view Clustering), Language Model (BERT, T5, GPT), Neural Codec Models (SoundStream, Encodec, AcademiCodec), Diffusion Models